Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Bartelink Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
12637 Van Allen RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 03/01/1983

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X130-XX13-XXXX

B. OPERATORS

Bartelink, Henry

Operator name: Bartelink, Henry

Telephone no.: (209) 838-3916 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12475 S Van Allen RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Bartelink, Henry

Legal owner name: Bartelink, Henry

Telephone no.: (209) 838-3916 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12475 S Van Allen RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	300	250	100	90
Number under roof	785	75	0	0	0	0
Maximum number	785	75	300	250	100	90
Average number	785	75	300	250	100	90
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,715.61 *tons per reporting period*

Total nitrogen from manure: 343,683.08 *lbs per reporting period*

After ammonia losses (30% loss applied): 240,578.16 *lbs per reporting period*

Total phosphorus from manure: 56,782.85 *lbs per reporting period*

Total potassium from manure: 152,397.72 *lbs per reporting period*

Total salt from manure: 386,863.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,708,000 *gallons*

Total nitrogen generated: 56,903.65 *lbs*

Total phosphorus generated: 8,065.37 *lbs*

Total potassium generated: 87,188.09 *lbs*

Total salt generated: 497,665.76 *lbs*

	12,708,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,708,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal SSSJID	Ground water
IW 1	Ground water
IW 2	Ground water
IW 3	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/01/2023	Solid commercial fertilizer *Commercial Fertilizer 4-0-0 (Corn Starter)*	23.50 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
06/25/2023	Solid commercial fertilizer *Commercial Fertilizer UN32 (Corn)*	5.40 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	5,330.66	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	5,330.66	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Behind Lagoon	29	29	2	process wastewater	X205-X130-XX13-XXXX
Heifer	47	47	2	process wastewater	X205-X130-XX13-XXXX
Jerry's Field	71	71	2	manure	X205-X030-XX04-XXXX
Rice Field	37	37	2	manure	X205-X150-XX16-XXXX
Small	4	4	2	process wastewater	X205-X130-XX13-XXXX
The 45	47	47	2	process wastewater	X205-X130-XX12-XXXX X205-X130-XX13-XXXX
Totals for areas that were used for application	235	235	12		
Totals for areas that were not used for application					
Land application area totals	235	235	12		

B. CROPS AND HARVESTS

Behind Lagoon

Field name: Behind Lagoon

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	524.42 *ton*	As-is		70.0	6,600.00	1,770.00	23,200.00		7.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.08	238.70	64.02	839.07	865.84

Behind Lagoon

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	946.20 *ton*	As-is		70.0	4,200.00	550.00	8,360.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.63	274.07	35.89	545.53	2,094.69

Heifer

Field name: Heifer

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/14/2023	850.70 *ton*	As-is		70.0	6,300.00	1,890.00	22,400.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.10	228.06	68.42	810.88	1,476.96

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,532.20 *ton*	As-is		70.0	3,090.00	370.00	2,950.00		2.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.60	201.47	24.12	192.34	410.76

Jerry's Field

Field name: Jerry's Field

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 71 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	1,285.10 *ton*	As-is		70.0	4,230.00	1,380.00	17,400.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.10	153.13	49.96	629.88	1,531.26

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 71 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	2,314.60 *ton*	As-is		70.0	3,240.00	710.00	4,530.00		4.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.60	211.25	46.29	295.36	974.09

Rice Field

Field name: Rice Field

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	669.70 *ton*	As-is		70.0	3,090.00	1,080.00	12,500.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.10	111.86	39.10	452.50	1,259.76

Rice Field

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,206.20 *ton*	As-is		70.0	3,360.00	460.00	2,930.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.60	219.07	29.99	191.04	1,154.04

Small

Field name: Small

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	72.40 *ton*	As-is		70.0	6,600.00	1,770.00	23,200.00		7.98

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.10	238.92	64.07	839.84	866.63

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 4 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	130.40 *ton*	As-is		70.0	3,360.00	430.00	3,580.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.60	219.07	28.04	233.42	1,251.84

The 45

Field name: The 45

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	850.70 *ton*	As-is		70.0	2,520.00	840.00	9,060.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.10	91.22	30.41	327.97	1,162.02

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,532.20 *ton*	As-is		70.0	3,870.00	850.00	5,080.00		5.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.60	252.32	55.42	331.22	1,011.25

NUTRIENT BUDGET

A. LAND APPLICATIONS

Behind Lagoon - 10/15/2022: Oats, silage-soft dough

Field name: Behind Lagoon

Crop: Oats, silage-soft dough — Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	143.81	15.59	278.18	1,448.65	912,000.00 *gal*
Application event totals		143.81	15.59	278.18	1,448.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	135.36	32.87	208.01	1,082.89	960,000.00 *gal*
Application event totals		135.36	32.87	208.01	1,082.89	

Behind Lagoon - 06/23/2023: Corn, silage

Field name: Behind Lagoon

Crop: Corn, silage — Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.93	0.00	0.00	438.72	3,960,000.00 *gal*
Application event totals		9.93	0.00	0.00	438.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Behind Lagoon - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	6.32	0.00	0.00	279.18	2,520,000.00 *gal*
Application event totals		6.32	0.00	0.00	279.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	237.08	24.33	372.05	1,918.27	1,344,000.00 *gal*
Application event totals		237.08	24.33	372.05	1,918.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.25	0.00	0.00	408.80	3,690,000.00 *gal*
Application event totals		9.25	0.00	0.00	408.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.25	0.00	0.00	408.80	3,690,000.00 *gal*
Application event totals		9.25	0.00	0.00	408.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.02	0.00	0.00	398.83	3,600,000.00 *gal*
Application event totals		9.02	0.00	0.00	398.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	8.57	0.00	0.00	378.89	3,420,000.00 *gal*
Application event totals		8.57	0.00	0.00	378.89	

Behind Lagoon - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	8.12	0.00	0.00	358.95	3,240,000.00 *gal*
Application event totals		8.12	0.00	0.00	358.95	

Heifer - 10/01/2022: Oats, silage-soft dough

Field name: Heifer

Crop: Oats, silage-soft dough

Plant date: 10/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	149.45	16.20	289.09	1,505.42	1,536,000.00 *gal*
Application event totals		149.45	16.20	289.09	1,505.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.46	31.44	198.94	1,035.66	1,488,000.00 *gal*
Application event totals		129.46	31.44	198.94	1,035.66	

Heifer - 06/02/2023: Corn, silage

Field name: Heifer

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Heifer - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	201.68	16.90	202.90	1,939.05	2,304,000.00 *gal*
IW 2	Ground water	6.68	0.00	0.00	295.31	4,320,000.00 *gal*
Application event totals		208.36	16.90	202.90	2,234.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.74	0.00	0.00	430.66	6,300,000.00 *gal*
Application event totals		9.74	0.00	0.00	430.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.19	0.00	0.00	406.05	5,940,000.00 *gal*
Application event totals		9.19	0.00	0.00	406.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.19	0.00	0.00	406.05	5,940,000.00 *gal*
Application event totals		9.19	0.00	0.00	406.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	9.05	0.00	0.00	399.89	5,850,000.00 *gal*
Application event totals		9.05	0.00	0.00	399.89	

Heifer - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	8.91	0.00	0.00	393.74	5,760,000.00 *gal*
Application event totals		8.91	0.00	0.00	393.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	8.35	0.00	0.00	369.13	5,400,000.00 *gal*
Application event totals		8.35	0.00	0.00	369.13	

Jerry's Field - 10/16/2022: Oats, silage-soft dough

Field name: Jerry's Field

Crop: Oats, silage-soft dough Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	170.33	58.30	96.21	1,218.68	1,000.00 *ton*
Application event totals		170.33	58.30	96.21	1,218.68	

Jerry's Field - 06/01/2023: Corn, silage

Field name: Jerry's Field

Crop: Corn, silage Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Jerry's Field - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	249.15	66.29	296.00	2,388.61	1,380.00 *ton*
Application event totals		249.15	66.29	296.00	2,388.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	67.80	9,945,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	63.81	9,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	63.81	9,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.81	

Jerry's Field - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	63.81	9,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	61.15	8,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	58.49	8,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	58.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	57.16	8,385,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.16	

Rice Field - 10/15/2022: Oats, silage-soft dough

Field name: Rice Field

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	163.42	55.93	92.31	1,169.27	500.00 *ton*
Application event totals		163.42	55.93	92.31	1,169.27	

Rice Field - 06/02/2023: Corn, silage

Field name: Rice Field

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	256.37	68.21	304.58	2,457.84	740.00 *ton*
Application event totals		256.37	68.21	304.58	2,457.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	68.87	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	68.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	63.77	4,875,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	63.77	4,875,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.77	

Rice Field - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	61.22	4,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	61.22	4,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	61.22	4,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal SSSJID	Ground water	0.00	0.00	0.00	56.12	4,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.12	

Small - 10/15/2022: Oats, silage-soft dough

Field name: Small

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.75	11.90	212.30	1,105.55	96,000.00 *gal*
Application event totals		109.75	11.90	212.30	1,105.55	

Small - 10/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.21	35.75	226.22	1,177.65	144,000.00 *gal*
Application event totals		147.21	35.75	226.22	1,177.65	

Small - 06/02/2023: Corn, silage

Field name: Small

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	209.82	17.58	211.10	2,017.32	204,000.00 *gal*
IW 1	Ground water	9.26	0.00	0.00	470.81	382,500.00 *gal*
Application event totals		219.08	17.58	211.10	2,488.13	

06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	13.07	0.00	0.00	664.68	540,000.00 *gal*
Application event totals		13.07	0.00	0.00	664.68	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	13.07	0.00	0.00	664.68	540,000.00 *gal*
Application event totals		13.07	0.00	0.00	664.68	

Small - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	11.98	0.00	0.00	609.29	495,000.00 *gal*
Application event totals		11.98	0.00	0.00	609.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	10.89	0.00	0.00	553.90	450,000.00 *gal*
Application event totals		10.89	0.00	0.00	553.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	10.89	0.00	0.00	553.90	450,000.00 *gal*
Application event totals		10.89	0.00	0.00	553.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	10.89	0.00	0.00	553.90	450,000.00 *gal*
Application event totals		10.89	0.00	0.00	553.90	

The 45 - 10/16/2022: Oats, silage-soft dough

Field name: The 45

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	131.55	31.95	202.15	1,052.36	1,512,000.00 *gal*
Application event totals		131.55	31.95	202.15	1,052.36	

The 45 - 05/31/2023: Corn, silage

Field name: The 45

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	12.98	0.00	0.00	659.97	6,300,000.00 *gal*
Application event totals		12.98	0.00	0.00	659.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	12.42	0.00	0.00	631.68	6,030,000.00 *gal*
Application event totals		12.42	0.00	0.00	631.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	240.32	24.66	377.14	1,944.51	2,208,000.00 *gal*
IW 1	Ground water	8.53	0.00	0.00	433.69	4,140,000.00 *gal*
Application event totals		248.85	24.66	377.14	2,378.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	12.23	0.00	0.00	622.25	5,940,000.00 *gal*
Application event totals		12.23	0.00	0.00	622.25	

The 45 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	12.23	0.00	0.00	622.25	5,940,000.00 *gal*
Application event totals		12.23	0.00	0.00	622.25	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	10.94	0.00	0.00	556.26	5,310,000.00 *gal*
Application event totals		10.94	0.00	0.00	556.26	

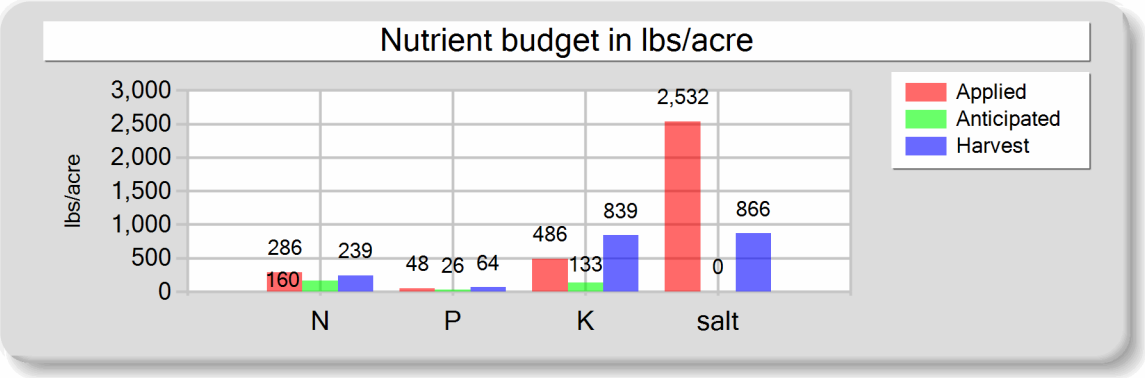
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	10.94	0.00	0.00	556.26	5,310,000.00 *gal*
Application event totals		10.94	0.00	0.00	556.26	

B. NUTRIENT BUDGET

Behind Lagoon - 10/15/2022: Oats, silage-soft dough

Field name: Behind Lagoon Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	279.18	48.46	486.20	2,531.54
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.18	48.46	486.20	2,531.54
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	238.70	64.02	839.07	865.84
Nutrient balance	47.47	-15.55	-352.88	1,665.70
Applied to removed ratio	1.20	0.76	0.58	2.92

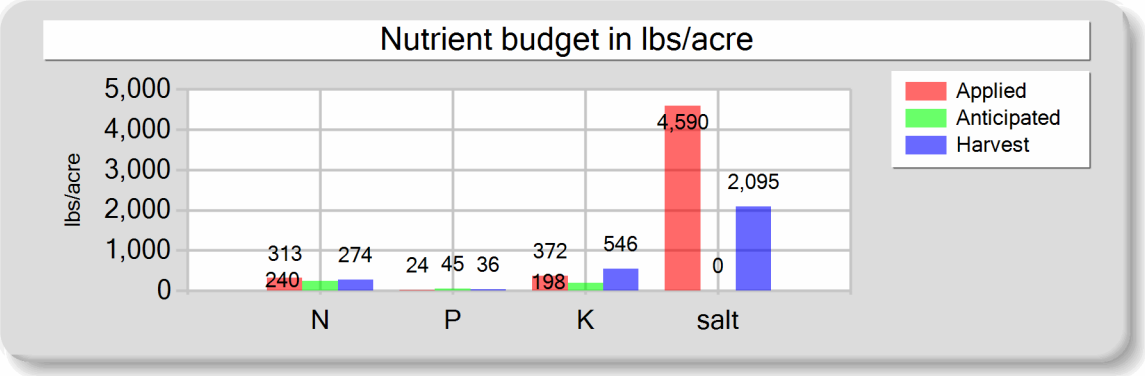
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,872,000.00 *gallons*
68.94 *acre-inches*
2.38 *inches/acre*

Total harvests for the crop
1 *harvests*

Behind Lagoon - 06/23/2023: Corn, silage

Field name: Behind Lagoon Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	237.08	24.33	372.05	1,918.27
Fresh water	60.45	0.00	0.00	2,672.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.53	24.33	372.05	4,590.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	274.07	35.89	545.53	2,094.69
Nutrient balance	38.46	-11.56	-173.48	2,495.76
Applied to removed ratio	1.14	0.68	0.68	2.19

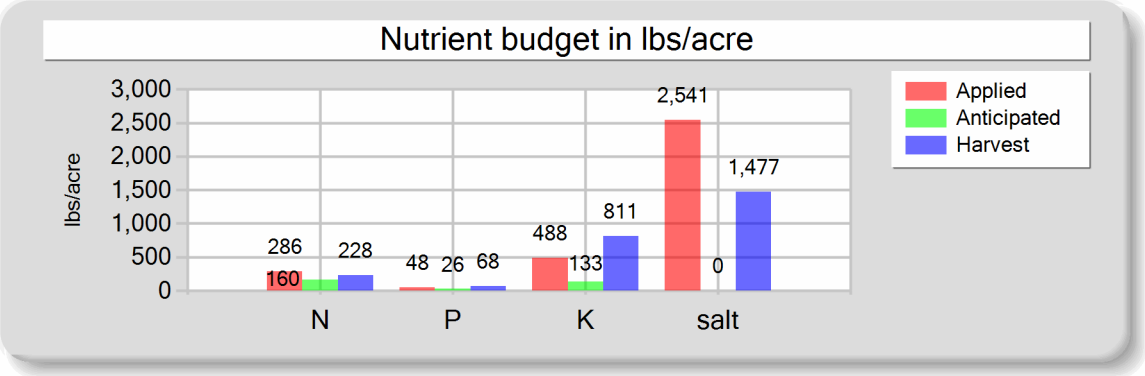
Fresh water applied	
24,120,000.00	*gallons*
888.26	*acre-inches*
30.63	*inches/acre*

Process wastewater applied	
1,344,000.00	*gallons*
49.49	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer - 10/01/2022: Oats, silage-soft dough

Field name: Heifer | Crop: Oats, silage-soft dough | Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	278.91	47.64	488.03	2,541.08
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.91	47.64	488.03	2,541.08
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	228.06	68.42	810.88	1,476.96
Nutrient balance	57.85	-20.78	-322.85	1,064.12
Applied to removed ratio	1.25	0.70	0.60	1.72

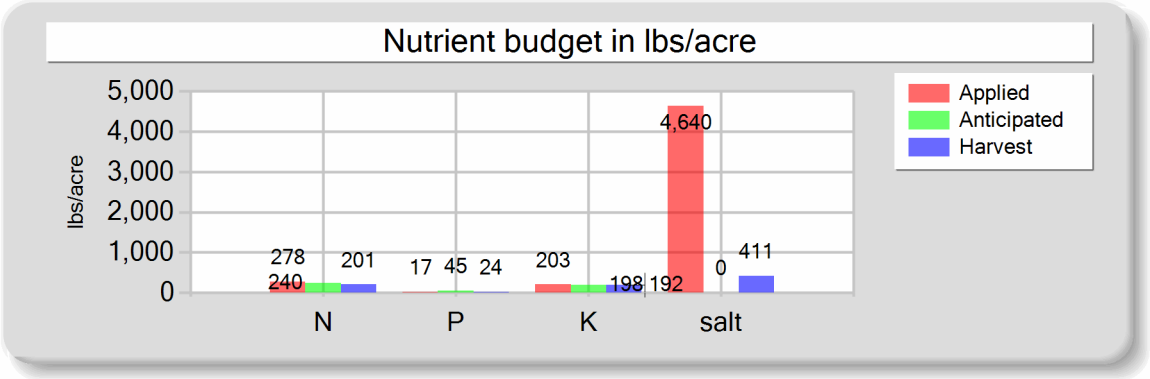
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
3,024,000.00	*gallons*
111.36	*acre-inches*
2.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Heifer - 06/02/2023: Corn, silage

Field name: Heifer | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.68	16.90	202.90	1,939.05
Fresh water	61.10	0.00	0.00	2,700.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.78	16.90	202.90	4,639.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.47	24.12	192.34	410.76
Nutrient balance	76.31	-7.23	10.56	4,229.12
Applied to removed ratio	1.38	0.70	1.05	11.30

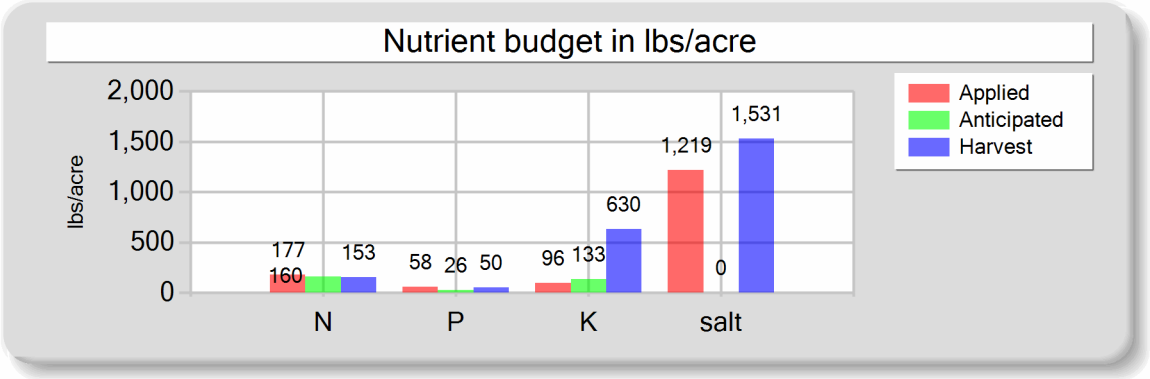
Fresh water applied	
39,510,000.00	*gallons*
1,455.02	*acre-inches*
30.96	*inches/acre*

Process wastewater applied	
2,304,000.00	*gallons*
84.85	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jerry's Field - 10/16/2022: Oats, silage-soft dough

Field name: Jerry's Field Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	170.33	58.30	96.21	1,218.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	177.33	58.30	96.21	1,218.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.13	49.96	629.88	1,531.26
Nutrient balance	24.20	8.34	-533.67	-312.58
Applied to removed ratio	1.16	1.17	0.15	0.80

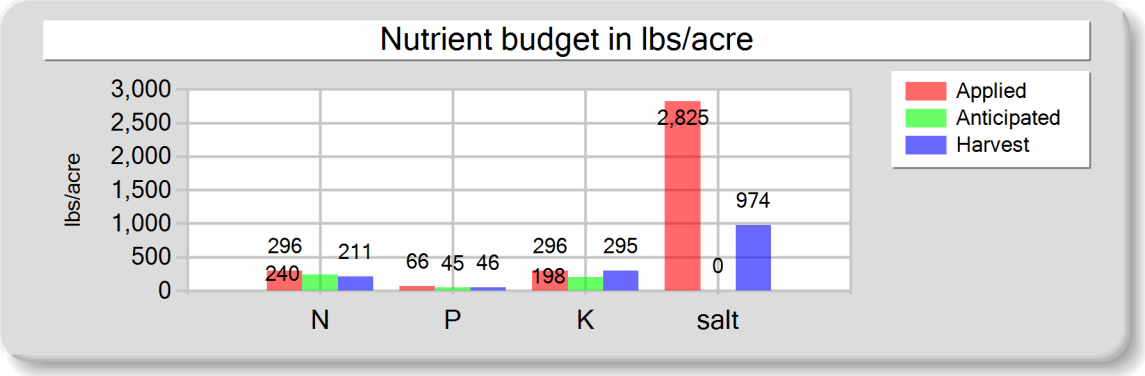
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Jerry's Field - 06/01/2023: Corn, silage

Field name: Jerry's Field Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	249.15	66.29	296.00	2,388.61
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	436.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.15	66.29	296.00	2,824.62
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	211.25	46.29	295.36	974.09
Nutrient balance	84.90	19.99	0.65	1,850.54
Applied to removed ratio	1.40	1.43	1.00	2.90

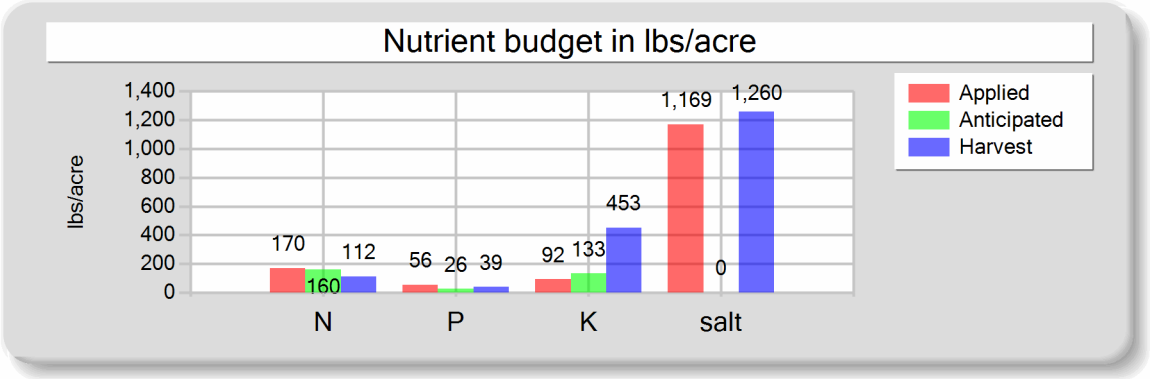
Fresh water applied	
63,960,000.00	*gallons*
2,355.43	*acre-inches*
33.18	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rice Field - 10/15/2022: Oats, silage-soft dough

Field name: Rice Field | Crop: Oats, silage-soft dough | Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.42	55.93	92.31	1,169.27
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.42	55.93	92.31	1,169.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	111.86	39.10	452.50	1,259.76
Nutrient balance	58.57	16.84	-360.19	-90.49
Applied to removed ratio	1.52	1.43	0.20	0.93

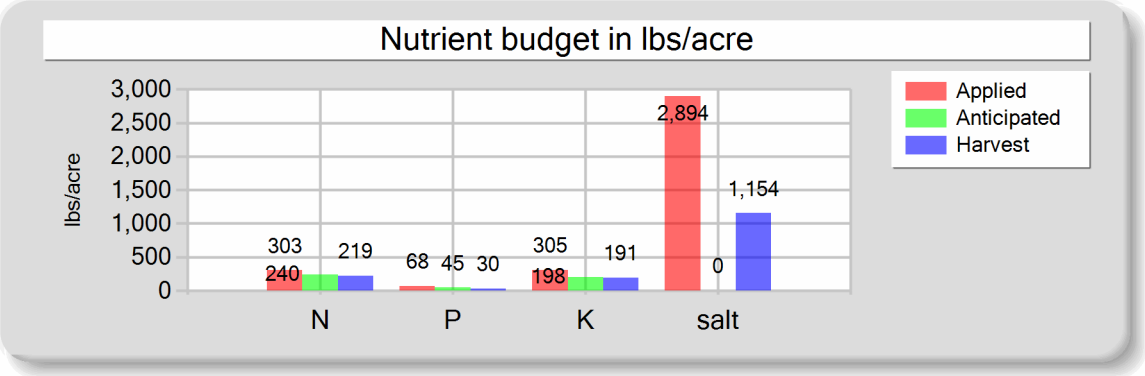
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Rice Field - 06/02/2023: Corn, silage

Field name: Rice Field Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	256.37	68.21	304.58	2,457.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	436.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	303.37	68.21	304.58	2,894.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.07	29.99	191.04	1,154.04
Nutrient balance	84.30	38.22	113.55	1,740.00
Applied to removed ratio	1.38	2.27	1.59	2.51

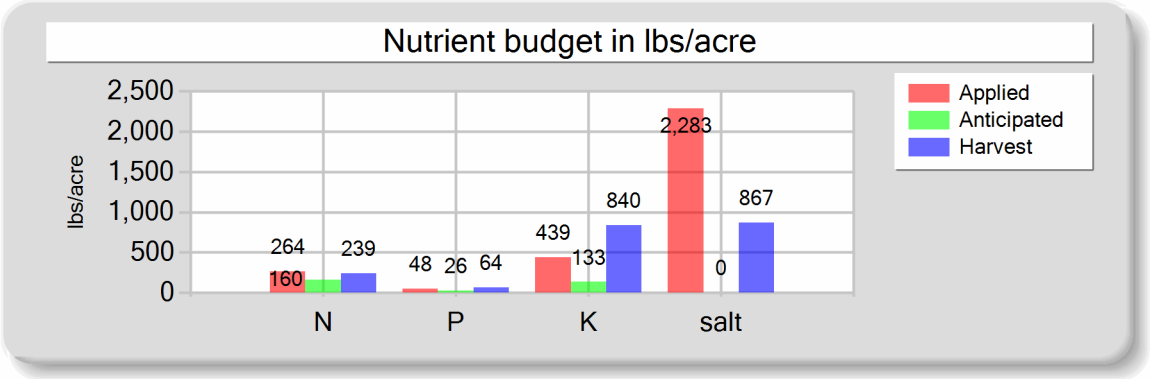
Fresh water applied	
33,345,000.00	*gallons*
1,227.98	*acre-inches*
33.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Small - 10/15/2022: Oats, silage-soft dough

Field name: Small | Crop: Oats, silage-soft dough | Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	256.96	47.65	438.51	2,283.19
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.96	47.65	438.51	2,283.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	238.92	64.07	839.84	866.63
Nutrient balance	25.04	-16.43	-401.33	1,416.56
Applied to removed ratio	1.10	0.74	0.52	2.63

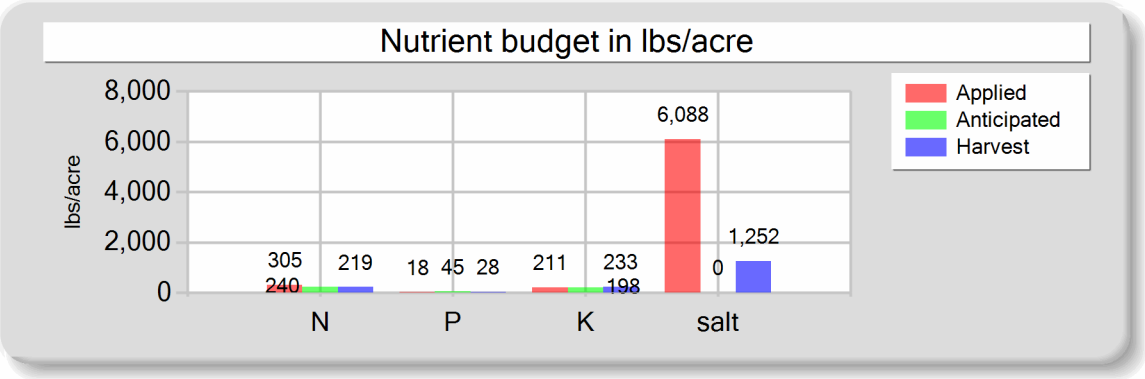
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
240,000.00	*gallons*
8.84	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

Small - 06/02/2023: Corn, silage

Field name: Small | Crop: Corn, silage | Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.82	17.58	211.10	2,017.32
Fresh water	80.04	0.00	0.00	4,071.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.86	17.58	211.10	6,088.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.07	28.04	233.42	1,251.84
Nutrient balance	85.79	-10.46	-22.32	4,836.64
Applied to removed ratio	1.39	0.63	0.90	4.86

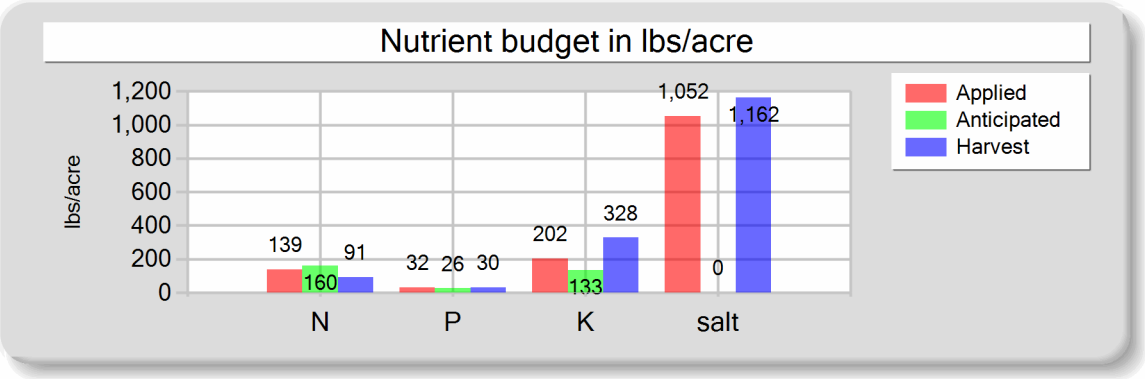
Fresh water applied	
3,307,500.00	*gallons*
121.80	*acre-inches*
30.45	*inches/acre*

Process wastewater applied	
204,000.00	*gallons*
7.51	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 45 - 10/16/2022: Oats, silage-soft dough

Field name: The 45 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	131.55	31.95	202.15	1,052.36
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	138.55	31.95	202.15	1,052.36
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	91.22	30.41	327.97	1,162.02
Nutrient balance	47.32	1.54	-125.82	-109.66
Applied to removed ratio	1.52	1.05	0.62	0.91

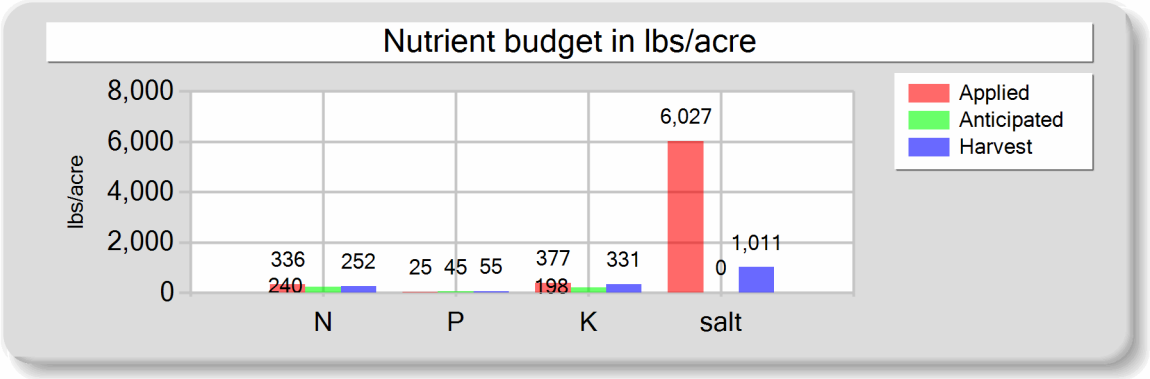
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,512,000.00	*gallons*
55.68	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 45 - 05/31/2023: Corn, silage

Field name: The 45 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.32	24.66	377.14	1,944.51
Fresh water	80.26	0.00	0.00	4,082.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.58	24.66	377.14	6,026.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	252.32	55.42	331.22	1,011.25
Nutrient balance	83.26	-30.76	45.92	5,015.61
Applied to removed ratio	1.33	0.44	1.14	5.96

Fresh water applied	
38,970,000.00	*gallons*
1,435.13	*acre-inches*
30.53	*inches/acre*

Process wastewater applied	
2,208,000.00	*gallons*
81.31	*acre-inches*
1.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Solids

Sample and source description: Separator Solids

Sample date: 09/12/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,900.00	8,180.00	13,500.00							17.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,100.00	3,900.00	27,700.00							79.90
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/17/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,800.00	5,800.00	25,900.00							20.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	6,150.00	45,500.00							21.60
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/09/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	4,280.00	8,990.00							8.84
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	548.00	385.00	0.00	0.00	59.40	1,060.00								7,490.00	5,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	490.00	252.00	0.00	0.00	119.00	753.00								5,290.00	3,920
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	493.00	199.00	0.00	0.00	41.30	496.00								5.64	4,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	613.00	426.00	0.00	0.00	62.90	962.00								7,790.00	4,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	286.00	174.00	0.00	0.00	82.70	678.00								4,540.00	3,400
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal SSSJID

Fresh Water

Sample description: Fresh Water

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								90.00	58
DL	0.50	0.50	0.50								1.00	10

IW 1

Fresh Water

Sample description: Fresh Water

Sample date: 09/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.60	0.00	11.60								751.00	590
DL	0.50	0.50	0.50								1.00	10

IW 2

Fresh Water

Sample description: Fresh Water

Sample date: 09/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.71	0.00	8.10								487.00	385
DL	0.50	0.50	0.50								1.00	10

IW 3

Fresh Water

Sample description: Fresh Water

Sample date: 09/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	23.20	0.00	22.60								1,120.00	800
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

Behind Lagoon

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

Heifer

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			90.00				
DL			4.00				

Jerry's Field

Jerry's Field

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			66.00				
DL			4.00				

Rice Field

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			94.00				
DL			4.00				

Small

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

The 45

The 45

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Behind Lagoon - 10/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,600.00	1,770.00	23,200.00		7.98
DL	100.00	100.00	100.00		0.01

Behind Lagoon - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	550.00	8,360.00		10.70
DL	100.00	100.00	100.00		0.01

Heifer - 10/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,860.00	1,160.00	13,800.00		13.60
DL	100.00	100.00	100.00		0.01

Heifer - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,090.00	370.00	2,950.00		2.10
DL	100.00	100.00	100.00		0.01

Jerry's Field - 10/16/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,230.00	1,380.00	17,400.00		14.10
DL	100.00	100.00	100.00		0.01

Jerry's Field - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,240.00	710.00	4,530.00		4.98
DL	100.00	100.00	100.00		0.01

Rice Field - 10/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,090.00	1,080.00	12,500.00		11.60
DL	100.00	100.00	100.00		0.01

Rice Field - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,360.00	460.00	2,930.00		5.90
DL	100.00	100.00	100.00		0.01

Small - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,360.00	430.00	3,580.00		6.40
DL	100.00	100.00	100.00		0.01

The 45 - 10/16/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,520.00	840.00	9,060.00		10.70
DL	100.00	100.00	100.00		0.01

The 45 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	870.00	5,150.00		5.17
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

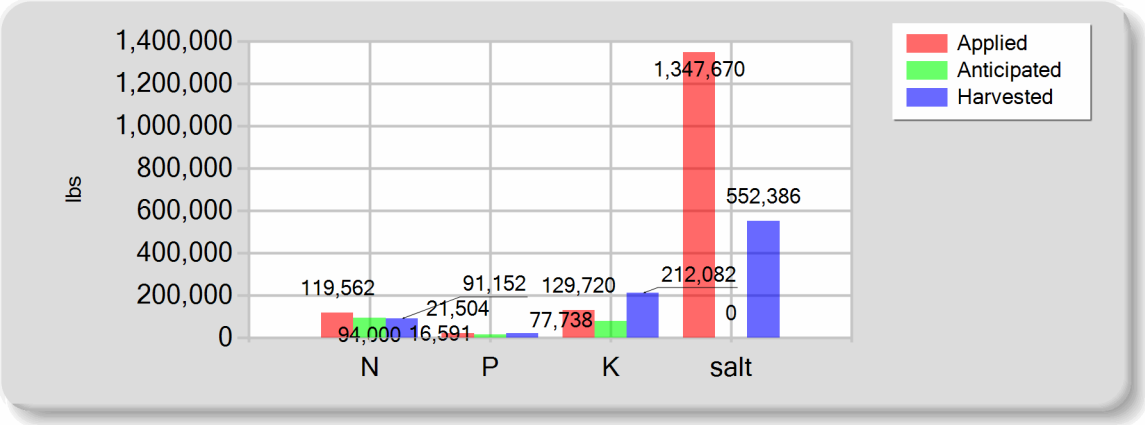
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

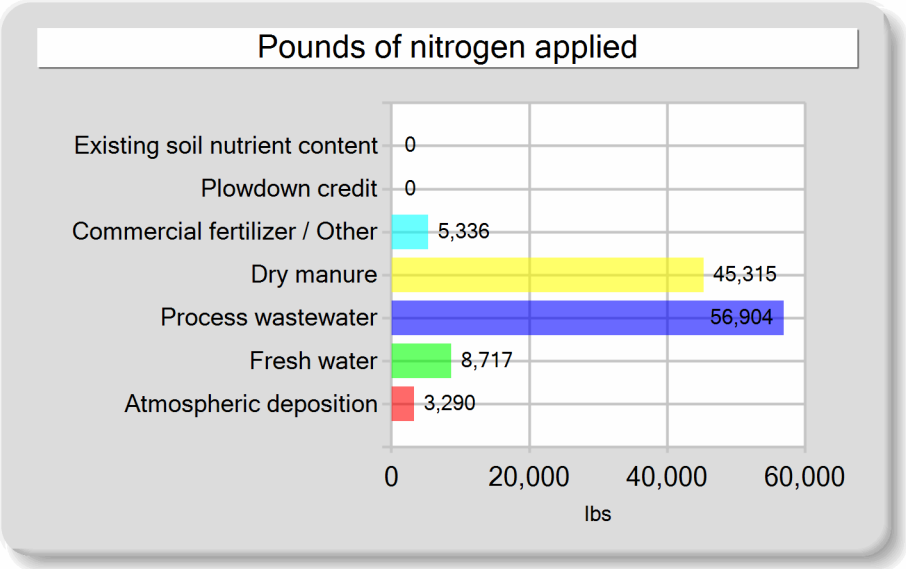
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

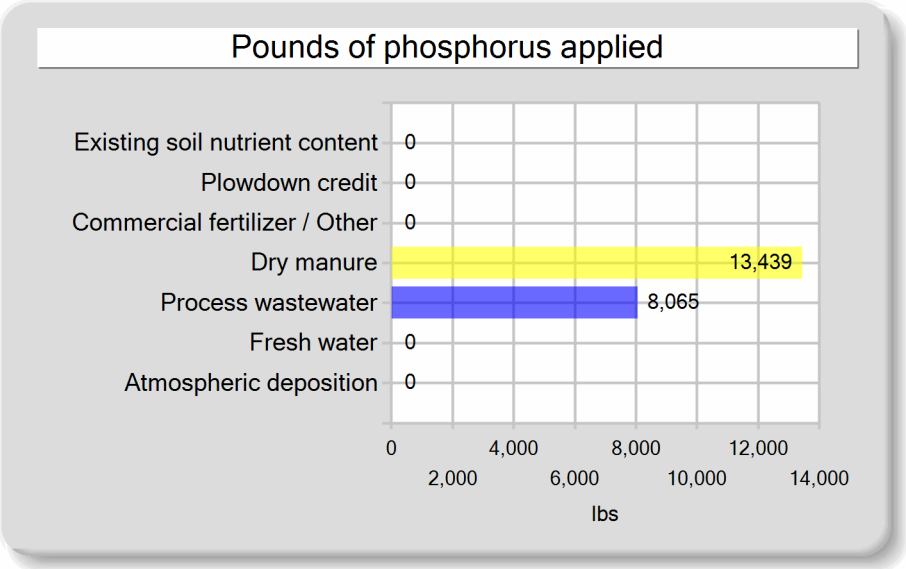
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,336.00	0.00	0.00	0.00
Dry manure	45,315.11	13,438.67	42,532.40	390,320.04
Process wastewater	56,903.65	8,065.37	87,188.09	497,665.76
Fresh water	8,717.49	0.00	0.00	459,684.03
Atmospheric deposition	3,290.00	0.00	0.00	0.00
Total nutrients applied	119,562.25	21,504.03	129,720.50	1,347,669.83
Anticipated crop nutrient removal	94,000.00	16,591.00	77,738.00	0.00
Actual crop nutrient removal	91,151.93	21,038.96	212,082.34	552,386.07
Nutrient balance	28,410.32	465.07	-82,361.84	795,283.76
Applied to removed ratio	1.31	1.02	0.61	2.44

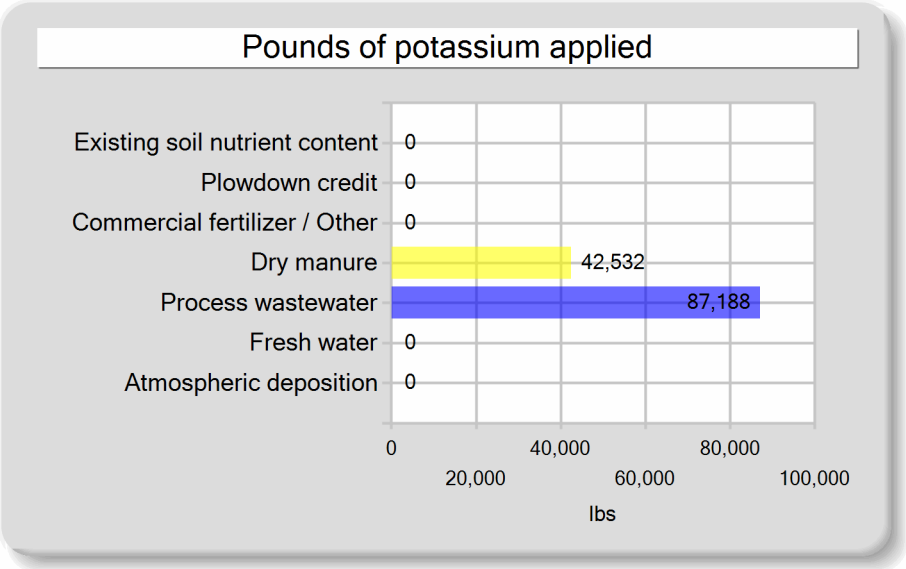
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

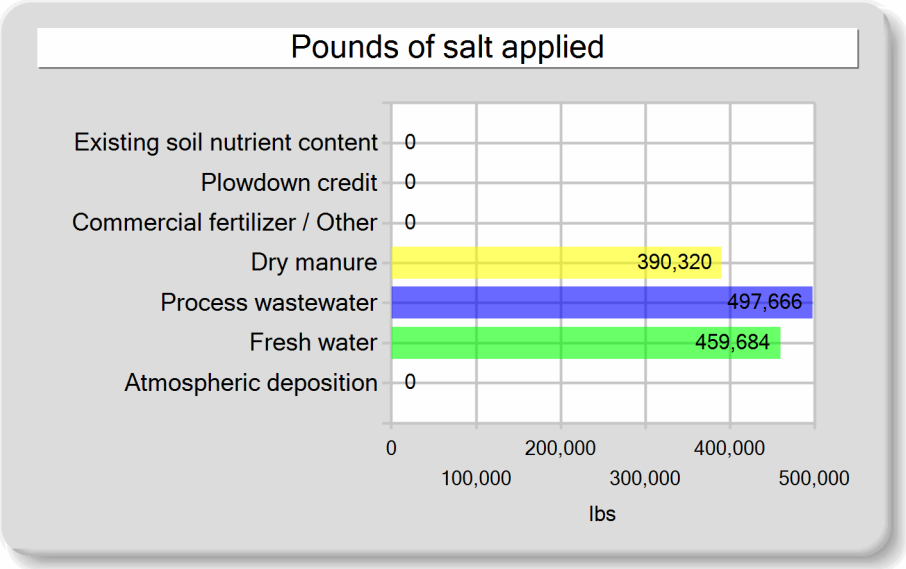


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Rice Field forage (1.52) and The 45 Field forage (1.52) both had high nitrogen field ratios as fields failed to meet anticipated nitrogen removal rate of 160 pounds/acre the subject fields removed 111.9 and 91.2 pounds of nitrogen/acre. Yields for both fields were adequate however a low nitrogen value from the plant tissue lab analysis resulted in only 6.18 and 5.04 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton. As a result, both fields had high nitrogen field ratios. Nitrogen application to the subject fields was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Henry Bartelink (May 29, 2024 11:54 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Henry Bartelink	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 29, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 11:24

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23I1834-01	IW 1	Ag Water			09/27/2023 7:00
23I1834-02	IW 2	Ag Water			09/27/2023 7:00
23I1834-03	IW 3	Ag Water			09/27/2023 7:00

Default Cooler Temperature on Receipt °C: 2.1
Containers Intact
COC/Labels Agree

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 11:24

Sample Results

Sample: IW 1
23I1834-01 (Water)

Sampled: 9/27/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.75**	mmhos/cm	0.01	1		09/29/23 13:03	SM 2510 B		BEI1078
Electrical Conductivity umhos	**751**	umhos/cm	10.0	1		09/29/23 13:03	SM 2510 B		BEI1078
Nitrate Nitrogen as NO3N	**11.6**	mg/L	0.1	1	10	09/29/23 22:46	EPA 300.0		BEI1029
Temperature	**25.0**	units	0.0	1		09/29/23 13:03	SM 4500-H+	H	BEI1078
pH	**8.1**	units	1.0	1		09/29/23 13:03	SM 4500-H+	H	BEI1078
Total Filterable Solids (TDS)	**590**	mg/L	10.0	1		10/03/23 13:23	SM 2540 C		BEJ0004
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/03/23 11:11	SM 4500-NH3 C		BEJ0001
Total Nitrogen	**11.6**	mg/L	1.00	1		10/03/23 11:11	SM 4500-NH3 C		BEJ0001

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 11:24

Sample Results
(Continued)

Sample: IW 2
23I1834-02 (Water)

Sampled: 9/27/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.49**	mmhos/cm	0.01	1		09/29/23 13:05	SM 2510 B		BEI1078
Electrical Conductivity umhos	**487**	umhos/cm	10.0	1		09/29/23 13:05	SM 2510 B		BEI1078
Nitrate Nitrogen as NO3N	**8.1**	mg/L	0.1	1	10	09/29/23 23:07	EPA 300.0		BEI1029
Temperature	**25.0**	units	0.0	1		09/29/23 13:05	SM 4500-H+	H	BEI1078
pH	**8.0**	units	1.0	1		09/29/23 13:05	SM 4500-H+	H	BEI1078
Total Filterable Solids (TDS)	**385**	mg/L	10.0	1		10/03/23 13:23	SM 2540 C		BEJ0004
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/03/23 11:12	SM 4500-NH3 C		BEJ0001
Total Nitrogen	**8.71**	mg/L	1.00	1		10/03/23 11:12	SM 4500-NH3 C		BEJ0001

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/04/2023 11:24

Sample Results
(Continued)

Sample: IW 3
23I1834-03 (Water)

Sampled: 9/27/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.12**	mmhos/cm	0.01	1		09/29/23 13:06	SM 2510 B		BEI1078
Electrical Conductivity umhos	**1120**	umhos/cm	10.0	1		09/29/23 13:06	SM 2510 B		BEI1078
Nitrate Nitrogen as NO3N	**22.6**	mg/L	0.1	1	10	09/29/23 23:28	EPA 300.0		BEI1029
Temperature	**25.0**	units	0.0	1		09/29/23 13:06	SM 4500-H+	H	BEI1078
pH	**8.2**	units	1.0	1		09/29/23 13:06	SM 4500-H+	H	BEI1078
Total Filterable Solids (TDS)	**800**	mg/L	10.0	1		10/03/23 13:23	SM 2540 C		BEJ0004
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/03/23 11:14	SM 4500-NH3 C		BEJ0001
Total Nitrogen	**23.2**	mg/L	1.00	1		10/03/23 11:14	SM 4500-NH3 C		BEJ0001

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

WATER WORK REQUEST

26230

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No | Bill To: ~~Acct #~~ 0 | Cons # 13

No. Samples: 3 | No of Bottles: 6

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Name: Bartelink Dairy
Address: 12637 Van Allen Rd
City: Escalon State: CA Zip: 95320
Telephone: 209 838-3916 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCW1 ++N

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 1	9/27/23	Start 7AM	2.1	
2	2	↓		7.1	
3	3	↓		3.3	
9					
10					

09/28/23 15:15 23I1834

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	~~9-28-23 DLV 10:15~~	9-28-23 10:15	9-28-23 2:35
Third				
Fourth	JY	DL	9/28/23 15:15	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping

$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/09/2023 08:44

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23I1835-01	DW Henry's	Drinking Water			09/27/2023 7:45
23I1835-02	DW Jerry's	Drinking Water			09/27/2023 7:45
23I1835-03	DW John's	Drinking Water			09/27/2023 7:45
23I1835-04	DW Milk Barn	Drinking Water			09/27/2023 7:45

Default Cooler Temperature on Receipt °C: 8.8
Containers Intact
COC/Labels Agree

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/09/2023 08:44

Sample Results

Sample: DW Henry's
23I1835-01 (Water)

Sampled: 9/27/2023 7:45
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.61**	mmhos/cm	0.01	1		09/29/23 13:07	SM 2510 B		BEI1078
Electrical Conductivity umhos	**1610**	umhos/cm	10.0	1		09/29/23 13:07	SM 2510 B		BEI1078
Ammonia (as N)	*	mg/L	0.00	1		09/27/23 07:45	Field		BEJ0263
Nitrate Nitrogen as NO3N	**42.2**	mg/L	0.1	1	10	09/29/23 02:27	EPA 300.0		BEI1030
Temperature	**25.0**	units	0.0	1		09/29/23 13:07	SM 4500-H+	H	BEI1078
pH	**8.0**	units	1.0	1		09/29/23 13:07	SM 4500-H+	H	BEI1078
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/04/23 11:49	SM 4500-NH3 C		BEJ0059
Total Nitrogen	**42.2**	mg/L	1.00	1		10/04/23 11:49	SM 4500-NH3 C		BEJ0059

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/09/2023 08:44

Sample Results
(Continued)

Sample: DW Jerry's
23I1835-02 (Water)

Sampled: 9/27/2023 7:45
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.06**	mmhos/cm	0.01	1		09/29/23 13:09	SM 2510 B		BEI1078
Electrical Conductivity umhos	**1060**	umhos/cm	10.0	1		09/29/23 13:09	SM 2510 B		BEI1078
Ammonia (as N)	*	mg/L	0.00	1		09/27/23 07:45	Field		BEJ0263
Nitrate Nitrogen as NO3N	**11.8**	mg/L	0.1	1	10	09/29/23 02:47	EPA 300.0		BEI1030
Temperature	**25.0**	units	0.0	1		09/29/23 13:09	SM 4500-H+	H	BEI1078
pH	**8.3**	units	1.0	1		09/29/23 13:09	SM 4500-H+	H	BEI1078
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/04/23 11:50	SM 4500-NH3 C		BEJ0059
Total Nitrogen	**11.8**	mg/L	1.00	1		10/04/23 11:50	SM 4500-NH3 C		BEJ0059

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/09/2023 08:44

Sample Results
(Continued)

Sample: DW John's
23I1835-03 (Water)

Sampled: 9/27/2023 7:45
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.28**	mmhos/cm	0.01	1		09/29/23 13:10	SM 2510 B		BEI1078
Electrical Conductivity umhos	**280**	umhos/cm	10.0	1		09/29/23 13:10	SM 2510 B		BEI1078
Ammonia (as N)	*	mg/L	0.00	1		09/27/23 07:45	Field		BEJ0263
Nitrate Nitrogen as NO3N	**2.6**	mg/L	0.1	1	10	09/29/23 03:07	EPA 300.0		BEI1030
Temperature	**25.0**	units	0.0	1		09/29/23 13:10	SM 4500-H+	H	BEI1078
pH	**8.0**	units	1.0	1		09/29/23 13:10	SM 4500-H+	H	BEI1078
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/04/23 11:51	SM 4500-NH3 C		BEJ0059
Total Nitrogen	**2.73**	mg/L	1.00	1		10/04/23 11:51	SM 4500-NH3 C		BEJ0059

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Bartelink Dairy
12637 Van Allen Rd
Escalon, CA 95320

Account# 00-0026230
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 09/28/2023 15:15
Reported: 10/09/2023 08:44

Sample Results
(Continued)

Sample: DW Milk Barn
23I1835-04 (Water)

Sampled: 9/27/2023 7:45
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.23**	mmhos/cm	0.01	1		09/29/23 13:18	SM 2510 B		BEI1078
Electrical Conductivity umhos	**228**	umhos/cm	10.0	1		09/29/23 13:18	SM 2510 B		BEI1078
Ammonia (as N)	*	mg/L	0.00	1		09/27/23 07:45	Field		BEJ0263
Nitrate Nitrogen as NO3N	**1.9**	mg/L	0.1	1	10	09/29/23 03:28	EPA 300.0		BEI1030
Temperature	**25.0**	units	0.0	1		09/29/23 13:18	SM 4500-H+	H	BEI1078
pH	**7.3**	units	1.0	1		09/29/23 13:18	SM 4500-H+	H	BEI1078
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/04/23 11:53	SM 4500-NH3 C		BEJ0059
Total Nitrogen	**2.00**	mg/L	1.00	1		10/04/23 11:53	SM 4500-NH3 C		BEJ0059

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

26230

Purchase Order No: | Bill To: Acct # 0 | Cons # 13

No. Samples: 4 | No of Bottles: 8

Results Need By:

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: Bastelink Dairy
Address: 12637 Van Allen Rd
City: Escalon State: CA Zip: 95320
Telephone: 209 838-3916 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (Please indicate Analysis)

DWW1 TTU

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

NO3 0 mg/L
42.2
11.9
206
1.9

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW Henry's	9/27/23	start 7:45 AM	8.8	
2	Jerry's	↓		5.8	
3	John's			8.5	
4	Milk Barn			5.3	
5					
6					
7					
8					
9					
10					

09/28/23 15:15 23I1835

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

Kara 9/29/23 (misplaced phone #)
10:48 - emailed Jeanette asking to call
10/2/23 14:45 - emailed J again)
10:42 - per Jeanette, ok & tnx

JG

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JCF	DLL	9-28-23 10:15	9-28-23 2:35
Third				
Fourth		DLL	9/28/23 15:15	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

2023 Annual Report Bartelink Dairy

Final Audit Report 2024-05-29

Created:	2024-05-29
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAmPuxDRg_aXp4JTPsRQBBxa2ubqa_d0fs

"2023 Annual Report Bartelink Dairy" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-05-29 - 3:00:45 PM GMT

Document emailed to bartelinkdairy@gmail.com for signature
2024-05-29 - 3:02:06 PM GMT

Email viewed by bartelinkdairy@gmail.com
2024-05-29 - 3:02:12 PM GMT

Signer bartelinkdairy@gmail.com entered name at signing as Henry Bartelink
2024-05-29 - 6:54:12 PM GMT

Document e-signed by Henry Bartelink (bartelinkdairy@gmail.com)
Signature Date: 2024-05-29 - 6:54:14 PM GMT - Time Source: server

Agreement completed.
2024-05-29 - 6:54:14 PM GMT

Adobe Acrobat Sign